Quantum Space-Time Bend of Special Relativity in Riemann Multidimensional Space Derivation Companion — TauTimp (Temporal Undulatory Theory) Project **Author:** Viorel-Mihai Popescu (TauTimp Project) **Source paper:** *Quantum Space-Time Bend of Special Relativity in Riemann Multidimensional Space*, 14May 2026 **Computational support:** Maple 2025 worksheet QmSp-TBend.mpl (Physics package) **This document:** English derivation companion — analytic steps, LaTeX equations, normalization constraints, and the link to the canonical TauTimp wave equation.

Table of Contents

1. Scope and Notation
2. The Onduscular Space Wave Equation
3. Separation of Variables and the Airy/Bessel Reduction
4. Boundary Conditions and the Cosmological Scale Parameters
5. The Schwarzschild Metric and Einstein's Field Equations
6. Coordinate Transformations and the Metric Variants
7. The Torsion Particle and the Second Field-Equation Variant
8. Attraction/Repulsion Transition and Observational Anomalies
9. Consistency with the Canonical TauTimp Framework
10. Summary of Key Results
11. References

1. Scope and Notation

This companion records the analytic derivations underlying the paper *Quantum Space-Time Bend of Special Relativity in Riemann Multidimensional Space*. The physical thesis is that spacetime curvature is not purely a geometric property of a pseudo-Riemannian manifold but emerges at the quantum level from mediating microparticles, one per physical coordinate of a generalized Riemannian multidimensional space.

Symbol conventions used throughout:

Symbol	Meaning
$f(r,t)$	Effective Onduscular Wave Function (EOWF) of space
R	TauTimp characteristic cosmological radius, $R = 3.9228 \times 10^{22}$ m
R_S	Muon-neutrino cosmological scale parameter (Schwarzschild-analogue for the Local Universe)
r	Radial coordinate
c	Speed of light in vacuum, $c = 299\,792\,458$ m/s
G	Newtonian gravitational constant, $G = 6.6743 \times 10^{-11}$ m^3 kg^{-1} s^{-2}
Δ	Laplacian operator
$\nu,\ \mu,\ \tau$	Electronic, muon, and tau neutrinos (mediators of gravity, space, and time)
P_{To}	Torsion particle (cylindrical-symmetry mediator)
P_γ	Gamma-ray particle
$m = M_\odot$	Solar mass, 1.9885×10^{30} kg

The metric signature is $(-,+,+,+)$, following the Misner–Thorne–Wheeler (MTW) convention. Coordinates are ordered $(x^1, x^2, x^3, x^4) = (ct, r, \theta, \phi)$.

2. The Onduscular Space Wave Equation 2.1 Muon-Neutrino Space Equation — Eq. (1)

The fundamental equation governing quantum space, mediated by the muon neutrino μ, is the variable-coefficient spherical wave equation

$$\frac{R_S}{r}\,\Delta f(r,t) = \frac{1}{c^2}\,\frac{\partial^2 f(r,t)}{\partial t^2}.$$

The coefficient R_S/r replaces the constant unity of the ordinary d'Alembert wave equation, introducing a position-dependent effective medium. In spherical symmetry the Laplacian is

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\,\frac{\partial f}{\partial r}.$$

2.2 Gravitational (Electronic-Neutrino) Equation — Eq. (3)

The corresponding equation for the gravitational field, mediated by the electronic neutrino ν (the graviton candidate), is structurally identical but uses the cosmological radius R:

$$\frac{R}{r}\,\Delta f(r,t) = \frac{1}{c^2}\,\frac{\partial^2 f(r,t)}{\partial t^2}.$$

2.3 Scaling Relationship Between (1) and (3)

The two equations are connected by the ratio

$$\frac{R_S}{R} = 1979.8.$$

Performing the substitution $r \to r/1979.8$ in Eq. (3) maps it onto Eq. (1):

$$\frac{R}{r/1979.8}\,\Delta f = 1979.8\,\frac{R}{r}\,\Delta f = \frac{R_S}{r}\,\Delta f.$$

Hence the EOWF of space is *proportional but not equal* to the EOWF of the gravitational field. The curvature of space is approximately 1980 times smaller than the curvature produced by the gravitational field — the structural distinction between the μ (space) and ν (gravity) sectors.

3. Separation of Variables and the Airy/Bessel Reduction 3.1 Separation

Insert $f(r,t) = g(r)\,u(t)$ into Eq. (3). Dividing by $g(r)\,u(t)$,

$$\frac{R}{r}\frac{1}{g}\left(g'' + \frac{2}{r}g'\right) = \frac{1}{c^2}\frac{\ddot{u}}{u} = -k^2,$$

with $k^2 > 0$ the separation constant. The temporal equation is the harmonic oscillator

$$\ddot{u}(t) + c^2k^2\,u(t) = 0 \quad \Rightarrow \quad u(t) = A\cos(ckt) + B\sin(ckt).$$

The radial equation is

$$g''(r) + \frac{2}{r}\,g'(r) + \frac{k^2}{R}\,r\,g(r) = 0.$$

3.2 Reduction via $g = \psi/r$

Substituting $g(r) = \psi(r)/r$ removes the first-derivative term:

$$g' = \frac{\psi'}{r} - \frac{\psi}{r^2}, \qquad g'' = \frac{\psi''}{r} - \frac{2\psi'}{r^2} + \frac{2\psi}{r^3},$$

so that $g'' + 2/r\,g' = \psi''/r$. Equation (3a) becomes the **Airy-type equation**

$$\psi''(r) + \frac{k^2}{R}\,r\,\psi(r) = 0.$$

3.3 Airy Form

Introduce the affine rescaling

$$\xi = -\left(\frac{k^2}{R}\right)^{1/3} r,$$

under which Eq. (3b) takes the canonical Airy form

$$\frac{d^2\psi}{d\xi^2} - \xi\,\psi = 0 \quad \Rightarrow \quad \psi(\xi) = c_1\,\mathrm{Ai}(\xi) + c_2\,\mathrm{Bi}(\xi).$$

Regularity of $g = \psi/r$ at the origin discards the Bi branch (and the part of Ai that does not vanish appropriately), leaving

$$g(r) = \frac{c_1}{r}\,\mathrm{Ai}\left[-(k^2/R)^{1/3}r\right].$$

3.4 Equivalent Bessel-1/3 Representation

Equation (3a) may instead be solved directly. With $g(r) = r^{-1/2}\,w(r)$ and $s = 2/3\,\sqrt{k^2/R}\;r^{3/2}$, the function w satisfies Bessel's equation of order $1/3$: $g(r) = r^{-1/2}\left[\,c_1 J_{1/3}(s) + c_2 J_{-1/3}(s)\,\right], \qquad s = \frac{2}{3}\sqrt{\frac{k^2}{R}}\;r^{3/2}.$

The Airy and Bessel-$1/3$ representations are identical, related by the standard identity $\mathrm{Ai}(-x) = 1/3\,\sqrt{x}\,[\,J_{1/3}(\zeta) + J_{-1/3}(\zeta)\,]$ with $\zeta = 2/3\,x^{3/2}$.

3.5 Eigenvalue Quantization

The boundary condition $g(2R_S) = 0$ (Section 4) selects a discrete spectrum. Writing a_n for the n-th negative zero of the Airy function, $\mathrm{Ai}(a_n) = 0$, the quantization condition is

$$-\left(\frac{k_n^2}{R}\right)^{1/3}(2R_S) = a_n \quad \Rightarrow \quad k_n^2 = R\,\frac{|a_n|^3}{(2R_S)^3}.$$

The associated temporal frequencies are $\omega_n = c\,k_n$, and the eigenfunctions are

$$f_n(r,t) = \frac{c_1}{r}\,\mathrm{Ai}\left[-(k_n^2/R)^{1/3}r\right]\left[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)\right].$$

4. Boundary Conditions and the Cosmological Scale Parameters 4.1 The Three Determining Conditions

Three independent conditions fix the basis of separated solutions:

Condition	Expression	Physical meaning
A	$f(2R_S,t) = 0$ for all t (equivalently $g(2R_S) = 0$)	The EOWF vanishes at the cosmological boundary $r = 2R_S$
B	$g'(0) = R$	The initial radial derivative at the origin equals the cosmological radius R
C	speed of interaction $= c$	Massless propagation, via the divergence condition Eq. (2)

4.2 Derivation of $2R_S$

The boundary radius is linked to the nuclear scale through the structural ratio of the nucleon and quark dimensions:

$$2R_S = R \cdot \frac{\dim(\text{nucleon})}{\dim(\text{quark})} = 2 \times 3.9228 \times \frac{0.841419}{0.43} \times 10^{25}\ \text{m}.$$

Numerically,

$$\frac{0.841419}{0.43} \approx 1.9568, \qquad 2R_S \approx 15.5329 \times 10^{25}\ \text{m} \approx 16.4187 \text{ billion light-years}.$$

The proton charge radius ≈ 0.8414 fm (CODATA) and the quark confinement radius ≈ 0.43 fm (QCD confinement scale) thus provide a sub-nuclear anchor for a cosmological-scale boundary — the structural bridge between the smallest and largest scales asserted by the framework.

5. The Schwarzschild Metric and Einstein's Field Equations 5.1 Einstein's Field Equation — Eq. (4)

$$R_{\mu\nu} - 1/2\ R_c\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = -\frac{8\pi G}{c^4}\, T_{\mu\nu} = -K\, T_{\mu\nu},$$

with the coupling constant

$$K = \frac{8\pi G}{c^4} = 2.076647 \times 10^{-43}\ \text{N}^{-1} \quad [\text{s}^2\,\text{m}^{-1}\,\text{kg}^{-1}].$$

Each term carries dimension $[\,L^{-2}\,]$. The sign convention $-K\,T_{\mu\nu}$ on the right-hand side follows MTW with signature $(-,+,+,+)$.

5.2 The Schwarzschild Metric Tensor

With $m = M_\odot$ and coordinates (ct, r, θ, ϕ),

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2Gm}{rc^2}\right) & 0 & 0 & 0 \\ 0 & -r^2 & 0 & 0 \\ 0 & 0 & -r^2\sin^2\theta & 0 \\ 0 & 0 & 0 & \left(1 - \frac{2Gm}{rc^2}\right)^{-1} \end{pmatrix}.$$

The line element — Eq. (5) — is

$$ds^2 = -\frac{rc^2 - 2Gm}{r}\, dt^2 + r^2\, d\theta^2 + r^2\sin^2\theta\, d\Phi^2 + \frac{rc^2}{rc^2 - 2Gm}\, dr^2,$$

with $e^{2\phi} = e^{-2\Lambda} = \left(1 - \frac{2Gm}{rc^2}\right)$, where ϕ and Λ are functions of r alone. Identifying the Schwarzschild radius $R_S = 2Gm/c^2$ recovers the textbook form

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2.$$

The vacuum solution satisfies $R_{\mu\nu} = 0$, $R_c = 0$, $\Lambda = 0$.

6. Coordinate Transformations and the Metric Variants 6.1 First Transformation (TransformCoordinates) — Eq. (5.1)

With the notation $R_{\text{Schwar}} = R_S$ and $R_S = 2Gm/c^2$, define the coordinate change

$$\text{TR:} \quad r \;\mapsto\; 1 - \frac{2\rho}{R_S} = \frac{mG - \rho c^2}{mG}.$$

Under TR the metric becomes

$$ds_{21}^2 = -\frac{c^6\,(mG - rc^2)}{m^2G^2\,(c^4r - 2mGc^2 + 2m^2G^2)}\, dr^2 + \frac{(mG - rc^2)^2}{m^2G^2}\, d\theta^2 + \frac{(mG - rc^2)^2}{m^2G^2}\sin^2\theta\, d\Phi^2 + \frac{c^4r - 2mGc^2 + 2m^2G^2}{mG - rc^2}\, dt^2.$$

For $T_{\mu\nu} = 0$ and $\Lambda = 0$ the field equation reduces to the identity $0 = 0$ — a coordinate transformation does not alter the physical content of the vacuum Schwarzschild solution. This is verified symbolically in the Maple worksheet.

6.2 Inverse Transformation — Eq. (6)

Applying the inverse change of variable $\rho = mG\,(1 - r)/c^2$ and re-expressing the metric as a function of r alone yields

$$ds_{22}^2 = -\frac{c^6 r}{m^2G^2\,(2mG - rc^2)}\, dr^2 + r^2\, d\theta^2 + r^2\sin^2\theta\, d\Phi^2 + \frac{2mG - rc^2}{r}\, dt^2.$$

For this metric the Ricci scalar and cosmological constant are **non-zero**:

$$R_{cp} = 2\,\frac{c^4 - G^2m^2}{c^4r^2}, \qquad \Lambda_p = \frac{c^4 - G^2m^2}{12\,c^4r^2}.$$

The modified metric tensor and proportional constant are

$$g_{\mu p} = \begin{pmatrix} \frac{rc^6}{G^2m^2(rc^2+2Gm)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin^2\theta & 0 \\ 0 & 0 & 0 & \frac{-rc^2-2Gm}{r} \end{pmatrix}, \qquad K_p = \frac{G}{c^2} = 7.42616 \times 10^{-28} \text{ [m/kg]}.$$

Metric (6) is *not* the standard Schwarzschild vacuum solution; it arises from a specific non-trivial change of variables. The condition $T_{\mu p} = 0$ requires $Gm = c^2$, i.e. $m = c^2/G \approx 1.35 \times 10^{27}$ kg, about 675 times smaller than $M_\odot$. The physical reading is that the γ-ray and torsion-field curvature is roughly 675 times smaller than the gravitational-field space curvature generated by the Sun.

6.3 Re-scaled Mass Substitution

With the substitution $m = 2M/R_S$ (M a re-scaled mass parameter), the cosmological constant becomes

$$\Lambda_\tau = \frac{c^4R_S^2 - 4G^2M^2}{12\,R_S^2\,c^4\,r^2},$$

and the corresponding metric tensor is

$$g_{\mu p} = \begin{pmatrix} \frac{-R_S^3c^6r}{4\,(-rR_Sc^2+4GM)\,G^2M^2} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin^2\theta & 0 \\ 0 & 0 & 0 & \frac{-rR_Sc^2+4GM}{rR_S} \end{pmatrix}.$$

The eigenvalues of the characteristic polynomial yield the closure condition $c^2 = mG$, which is exactly what the substitution $m = 2M/R_S$ enforces, producing the null tensor $T_{\mu p} = 0$.

7. The Torsion Particle and the Second Field-Equation Variant

The torsion particle P_{To} — associated with the synchronized precession of an ensemble of spin-bearing particles — introduces a second set of field equations with **cylindrical symmetry**:

$$R_{\mu p} - 1/2\; R_{cp}\, g_{\mu p} + \Lambda\, g_{\mu p} = -K_p\, T_{\mu p}, \qquad K_p = \frac{G}{c^2} = 7.42616 \times 10^{-28} \text{ [m/kg]}.$$

This parallels Einstein's Eq. (4) but with a distinct Ricci tensor $R_{\mu p}$, scalar R_{cp}, cosmological constant Λ_p, and coupling K_p (dimensionally [m/kg] rather than [N^{-1}]). The boundary condition is $f(2R, t) = 0$, the same form as the gravitational ν case, but applied in cylindrical rather than spherical geometry.

Side-by-side comparison:

Quantity	Gravity (ν)	Torsion (P_{To})
Coupling	$K = 8\pi G/c^4 = 2.077 \times 10^{-43}$ N^{-1}	$K_p = G/c^2 = 7.426 \times 10^{-28}$ m/kg
Symmetry	Spherical	Cylindrical
Ricci scalar	R_c (standard)	$R_{cp} = 2(c^4 - G^2m^2)/(c^4r^2)$
Cosmological constant	$\Lambda = 0$ (vacuum)	$\Lambda_p \neq 0$

Because each coordinate bends spacetime through a **first-order linear** equation, the total spacetime curvature is treated as **cumulative (superposable)** — the contributions of the ν, τ, μ, P_{To} and P_γ sectors add. This superposition holds in the linearized (weak-field) regime, the same regime in which gravitational-wave theory is formulated.

8. Attraction/Repulsion Transition and Observational Anomalies 8.1 Transition Scale

The γ-ray particle P_γ and the torsion particle P_{To} are proposed as the agents of cosmic expansion, with an attraction/repulsion transition at

$$2R \approx 8.293 \text{ million light-years} \approx 7.846 \times 10^{22} \text{ m}.$$

- For $r < 2R$: P_{To} generates **attraction** between galaxies.
- For $2R < r < 2R_S \approx 16.418$ billion light-years: P_{To} generates **repulsion**, playing the role of dark energy.

The transition scale is set directly by the canonical TauTimp radius $R = 3.9228 \times 10^{22}$ m, since $2R = 7.846 \times 10^{22}$ m.

8.2 Cited Observational Anomalies

The cumulative ($\nu + P_{To}$) bending is offered as the origin of three astronomical anomalies:

1. **Venus trajectory anomaly** — a cumulative-curvature effect known in astronomical circles.
2. **Halley's Comet perihelion delay (1910)** — a 3-day delay at perihelion, cited from Prof. Dr. Ioan N. Popescu, *Gravity* (1982), p. 184.
3. **Mercury perihelion advance** — observed at $42.6'' \pm 0.9''$ per century (Popescu, *Gravity*, p. 378), against the standard general-relativistic value of $\approx 43''$ per century.

9. Consistency with the Canonical TauTimp Framework

The canonical TauTimp wave equation is the variable-coefficient spherical PDE

$$\frac{R}{r}\,\Delta f = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}, \qquad R = 3.9228 \times 10^{22}\ \text{m},$$

which is exactly Eq. (3) of the present paper. The space-bend formulation thus inherits the canonical TauTimp infrastructure without modification:

- **Same operator structure** — the $(R/r)\Delta$ coefficient, separation of variables, and the Airy reduction $g = \psi/r$ are identical to the established TauTimp derivation pipeline.
- **Same eigenstates** — the Airy-function eigenstates with quantization from the zeros a_n of Ai carry over directly.
- **Same boundary radius** — $R = 3.9228 \times 10^{22}$ m (≈ 4.1 Mly) is the canonical TauTimp cosmological boundary parameter; $2R$ is the attraction/repulsion transition.
- **New parameter** — the muon-sector scale R_S, related by $R_S/R = 1979.8$, is specific to the space-bend paper. Equation (1) is the μ-sector counterpart of the canonical ν-sector Eq. (3), and the substitution $r \to r/1979.8$ maps one onto the other.

In short, Eq. (3) *is* the canonical TauTimp equation; Eq. (1) is its rescaled muon-sector partner; and the Schwarzschild/torsion analysis of Sections 5–7 supplies the general-relativistic embedding for the same framework.

10. Summary of Key Results

Result	Expression	Value
Characteristic radius	R	3.9228×10^{22} m
Muon-neutrino cosmological scale	R_S	$R_S = R \cdot 1979.8/2 \approx 3.882 \times 10^{25}$ m
Observable-universe boundary	$2R_S$	≈ 16.418 Gly
Spatial/gravitational curvature ratio	R_S/R	≈ 1979.8
Schwarzschild coupling	$K = 8\pi G/c^4$	2.0766×10^{-43} N^{-1}
Torsion coupling	$K_p = G/c^2$	7.4262×10^{-28} m/kg
Attraction/repulsion transition	$2R$	$\approx 7.846 \times 10^{22}$ m ≈ 8.29 Mly
Ricci scalar (metric 6)	R_{cp}	$2(c^4 - G^2m^2)/(c^4r^2)$
Cosmological constant (metric 6)	Λ_p	$(c^4 - G^2m^2)/(12\,c^4r^2)$
Re-scaled cosmological constant	Λ_τ	$(c^4R_S^2 - 4G^2M^2)/(12\,R_S^2c^4r^2)$
Radial eigenvalue spectrum	k_n^2	$R\,\lvert a_n\rvert^3/(2R_S)^3,\ \mathrm{Ai}(a_n) = 0$

11. References

[1] MIT, *Einstein's Field Equation — Schwarzschild Solution Derivation*, pp. 7–10. http://web.mit.edu/klmitch/classes/8.033/Schwarzschild.pdf

[2] C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation*, W. H. Freeman, 1973 (copyright 1970–1971).

[3] Prof. Dr. Ioan N. Popescu, *Gravity (Gravitația)*, Editura Științifică și Tehnică, Bucharest, 1982.

Computational links (author's site): - Schwarzschild AXR result: https://michaelvio.github.io/SchwarAXR.pdf - Sun metric: https://michaelvio.github.io/Sun.pdf - Proportional constant derivation: https://michaelvio.github.io/ConstProp.pdf - Tau bending: https://michaelvio.github.io/TauBend.pdf - Gravitational flux tau: https://michaelvio.github.io/GFtau.pdf - Gravity bending overview: https://michaelvio.github.io/GravBend.pdf - viXra preprint: https://vixra.org/pdf/1511.0101v1.pdf https://michaelvio.github.io/QTBendP.pdf

End of Derivation Companion — TauTimp Project.